EVIDENCE BEFORE INTERVENTION

HVAC Service | Companion Technical Reference | v1.0 | 16 October 2026

Purpose. A practical examination reference accompanying the eight-lesson TA14 HVAC showroom. It separates observed system reality, diagnostic reasoning, proposed action, authorization, bounded execution, and verified outcome.

Status: Editorial teaching and technical examination draft. Not an adopted Academy standard, field authorization, ASHRAE endorsement, or substitute for applicable laws, safety procedures, licensing, manufacturer requirements, and the controlling TA14 Academy materials.

The governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

The evidence-to-proof discipline

Preserve original conditions; collect non-invasive evidence; evaluate the applicable Non-Invasive Refrigerant Entry Threshold (NIRET); collect refrigerant-side evidence only when justified and separately authorized; establish the original-state HVAC Performance Record; declare the diagnostic determination; authorize and perform bounded intervention; capture the post-performance record; then compare records to support an outcome.

This is a conceptual sequence, not permission to connect gauges or alter equipment. The fourteen Academy inspection steps are a teaching route, not fourteen separate grants of execution authority.

Eight showroom lessons

#	Lesson	Question to answer
01	Preserve reality	What was present before anything changed?
02	Original-state performance record	What attributable baseline measurements and conditions were preserved?
03	Seven In / Seven Out	Was the full field examination followed and documented?
04	NIRET	Is refrigerant entry evidentially necessary under the current Academy standard?
05	Declared Diagnostic Determination	What conclusion is supported, and what remains uncertain?
06	Establish authority	Who may perform which work, where, and under what conditions?

#	Lesson	Question to answer
07	Post-performance record	What did the system do after the authorized intervention?
08	Proof of performance	What outcome is actually supported by comparable records?

01 / 02 PRESERVE REALITY AND BASELINE

Before changing settings or opening a consequential boundary

• Record the complaint in the customer's words; distinguish complaint from observed symptoms.

• Identify the equipment, system configuration, model and serial number, site, technician, date and time.

• Capture initial control settings, operating mode, visible condition, access limitations, and safety observations.

• Preserve original photographs, readings, notes and provenance. Record unavailable data explicitly rather than reconstructing it later.

Original-state HVAC Performance Record (HPR)

The HPR is an attributable record of the system as found. It should state the operating conditions under which measurements were taken and distinguish measured values from estimates or interpretations.

Evidence category	Representative fields / limitations
Identity and context	Equipment identity; technician; time; weather/ambient; mode; setpoints; run duration.
Airside	Return and supply temperatures; temperature difference; filter/coil condition; static pressure and airflow evidence when safely obtainable.
Electrical / controls	Observed sequence, voltage/current measurements where appropriate, controls, safeties, faults.
Refrigerant-side	Only if NIRET is met and separate authority established; record instrument, connection, condition and measurement provenance.
Integrity	Raw observations retained; later adjustments separately attributed; missing data, uncertainty and comparability limits identified.

Do not assume: A single temperature split, gauge reading, or photograph proves a fault or authorizes a repair.

Lesson transition

After preserving opening reality, the field examination tests competing explanations without silently changing the baseline.

03 SEVEN IN / SEVEN OUT

The following fourteen titles are drawn from the TA14 Exchange Academy field sequence. They are inspection and teaching checkpoints; specific access, electrical, cleaning, and refrigerant work remains subject to safety and authorization boundaries.

Step	Seven In - indoor examination
01	Approaching & Opening the Air Handler
02	Heat Kit & Furnace Operation Check
03	Blower Motor & Airflow Integrity
04	Evaporator Coil Inspection
05	Metering Device Identification
06	Filter & Return Airflow Check
07	Air Handler Reassembly & Final Indoor Verification

Step	Seven Out - outdoor examination
08	Condenser Visual Inspection & Controlled Panel Opening
09	Start Components, Capacitors & Compressor Insulation Verification
10	Compressor & Fan Motor Amp Draw Verification
11	Condenser Panel Reassembly & Mechanical Integrity Check
12	Condenser Coil Cleaning & Airflow Restoration
13	Refrigerant Charge Approximation
14	Condensate Management Verification & System Closure

Boundary note: Steps referring to opening panels, electrical testing, coil cleaning, or refrigerant approximation do not waive the need for safe conditions and applicable authority. Step 12 may involve an intervention; if cleaning would alter the baseline, preserve the original record and establish authorization first.

04 / 05 NIRET AND DIAGNOSTIC DETERMINATION

NIRET - necessity is not permission

Evaluate the current controlling Academy NIRET standard against preserved non-invasive evidence and unresolved diagnostic questions. A threshold finding must identify its evidence, rule version, uncertainty and result. Even a satisfied NIRET does not itself grant permission for refrigerant-side work.

Gate	Required disposition
Non-invasive evidence insufficient or NIRET not met	Do not enter the refrigerant system. Continue permitted investigation or HOLD / ESCALATE.
NIRET met; authority missing	HOLD. Establish applicable authority, actor standing, qualifications, safe conditions and scope.
NIRET met and authority established	Only the bounded, necessary refrigerant-side examination may proceed; preserve new observations.

Declared Diagnostic Determination (DDD)

A DDD is a recorded conclusion supported by evidence, not an intervention order. State observed facts, analysis and inference separately; cite the record; acknowledge alternatives, uncertainty, excluded causes and unresolved questions. The determination may be supported, limited, indeterminate or require additional authorized testing.

Diagnostic termination - unresolved source verification

The exact controlling Academy wording for a formal diagnostic-termination requirement has not been verified in this edition. This reference therefore does not invent a mandatory termination formula or equate termination with ALLOW, HOLD, DENY or ESCALATE. The controlling Academy source must be checked before governing release.

Illustrative example, not an actual diagnosis

A high head-pressure reading alongside a visibly obstructed condenser may support an airflow hypothesis, but does not by itself establish a definitive root cause. Any proposed cleaning, testing, replacement or refrigerant work still requires its own evidentiary and authorization basis.

06 ESTABLISH AUTHORITY BEFORE INTERVENTION

Diagnostic support and permission to act are separate. An intervention is eligible for execution only when its evidence, applicable authority and actor standing are established for this specific system, location, scope and moment.

Check	Minimum examination question
Evidence	Is the proposed work tied to preserved, admissible observations and a declared determination?
Applicable authority	Are customer/site permissions, codes, licensing and manufacturer requirements satisfied?
Established standing	Is the actor qualified and authorized in the relevant capacity and relationship?
Scope and limits	Are the equipment, permitted actions, exclusions, time, safety conditions and stop triggers explicit?
Execution record	Can the actual work be attributed, bounded, and distinguished from the proposal?
Reassessment	Will changed conditions, unexpected findings or scope expansion cause HOLD or ESCALATE?

Seven governance boundaries

Evidence -> Diagnostic Determination -> Proposed Intervention -> Intervention Authorization -> Bounded Execution -> Post-Intervention Verification -> Outcome.

These are governance distinctions overlaid on the field route; they do not replace the fourteen-step Academy inspection sequence.

Urgent protective action

An immediate safety response may justify narrowly bounded protective action under applicable law and site rules. Record the hazard, necessity, actor, authority, work performed, evidence lost, and limitations on later proof. Urgency does not erase accountability or authorize unrelated work.

Disposition discipline

ALLOW only within the proven scope; HOLD when prerequisites are missing; DENY where prohibited or unsupported; ESCALATE when higher authority or specialized evaluation is required.

07 / 08 POST-RECORD AND PROOF OF PERFORMANCE

Post-intervention HPR

Preserve a separate record after authorized work, including work actually performed, timing, new system configuration, operating mode, setpoints, ambient/load conditions, instruments, measurements, safety checks and any new unresolved observations.

Comparability before outcome

Do not treat before/after numbers as comparable unless the relevant operating conditions and measurement methods are sufficiently aligned or differences are explicitly accounted for. Preserve raw and adjusted values separately. State what cannot be concluded.

Outcome	Meaning
IMPROVED	Supported beneficial change relative to a defensible baseline.
RESTORED	Supported return to an established functional or performance criterion.
UNCHANGED	No supported material change under comparable conditions.
DEGRADED	Supported adverse change.
PARTIALLY RESOLVED	Some issues supported as resolved; others remain.
UNRESOLVED	The reported issue remains or resolution has not been established.
INDETERMINATE	Evidence or comparability is insufficient for a defensible outcome.

Critical failures to prevent

- Consequential intervention before original-state evidence is preserved.
- Refrigerant entry before applicable NIRET and separate authority.
- Fabricating, backfilling, reconstructing or silently altering baseline evidence.
- Failing to declare a diagnostic determination before ordinary intervention.
- Executing unsupported or unauthorized work, or exceeding the established scope.
- Claiming improvement from a post-state record alone.
- Unsafe or unlawful conduct, including ignoring urgent protective obligations.

REVIEW, LIMITATIONS AND SOURCE REGISTER

Review checklist for showroom publication

• Confirm the Academy's controlling NIRET text and version.

• Verify exact Google Sites Academy language concerning diagnostic termination.

• Reconcile every image caption with the source sequence and authority boundaries.

• Label all invented equipment identities, readings, dates, signatures and licenses as illustrative; do not present them as attested records.

• Correct illustrative slides that suggest gauge connection precedes NIRET, or that a completed post-record automatically proves improvement.

• Keep frozen D3/D08 material unchanged without separate authorization.

• Conduct independent technical/safety review before adopting as governing procedure.

Source basis and provenance

Primary internal reference: TA14 Exchange repository, HVAC Academy competency and assessment blueprint; fourteen-step HVAC Academy field lessons; Academy outcome-proof materials; and TA14 HVAC Technical Reconciliation v4.1 (editorial examination draft).

Public entry points: https://www.ta14exchange.com/academy/hvac and https://www.ta14exchange.com/academy/hvac/blueprint

Source limitation: This edition synthesizes the Academy framework and the proposed eight-image showroom teaching plan. It is not a verbatim reproduction of the controlling Google Sites field standard; diagnostic-termination wording and the complete current NIRET rule have not been independently verified here.

Release disposition

ALLOW: bounded educational examination and technical review. HOLD: governing publication, normative field adoption, and claims of institutional endorsement until source reconciliation and technical validation are complete.

TA14 AUTHORITY GOVERNANCE INSTITUTION | Evidence before intervention. Capability is not authority.